



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-2026)
Faculty Name	Ms. G. Mamatha
Student Name	Brundavanam VR VR SrisaiHarini
Roll Number	2520030443
Branch	CSE
Course Outcome	CO-1 Binary Search Tree, AVL Tree, Tree Traversal

Case Study Topic: LibraryNet – Intelligent Digital Library & Knowledge Retrieval System

Work Done Summary:

In this case study, we implemented a digital library management system named LibraryNet using various Data Structures and Algorithms. The system efficiently manages books, journals, digital resources, borrowing records, and library analytics.

We implemented Binary Search Trees and AVL Trees for efficient indexing and searching of books using BookID. Multiway trees such as B-Trees and B+ Trees were used for large database indexing and range-based searching. Graph algorithms like BFS and DFS were applied to explore related books and citation networks.

The project demonstrates efficient searching, insertion, deletion, recommendation, and analytics functionalities in a digital library environment.

Implementation Details:

- Binary Search Tree (BST)
- Used for storing books based on BookID.
- Supports efficient searching and insertion.
- Helps retrieve books quickly.
- AVL Tree
- Maintains balanced structure after insertion/deletion.
- Avoids skewed tree formation.
- Improves search efficiency.
- Tree Traversals
- Inorder Traversal → Displays books in sorted order.
- Preorder Traversal → Displays hierarchy.
- Postorder Traversal → Used for deletion operations.

RESULT/SCREENSHOT:

```
C:\Users\HARINI\OneDrive\Documents\JAVA PROGS>java BST
Book Found:
Book ID: 105
Title: Python
```

Time Complexity

Operation	Time Complexity
BST Search	$O(\log n)$ average
BST Insertion	$O(\log n)$ average
AVL Search	$O(\log n)$
AVL Insertion	$O(\log n)$
Tree Traversal	$O(n)$

GitHub Link: <https://github.com/HariniBrundavanam/DSA-2-project>

Student Signature	Faculty Signature